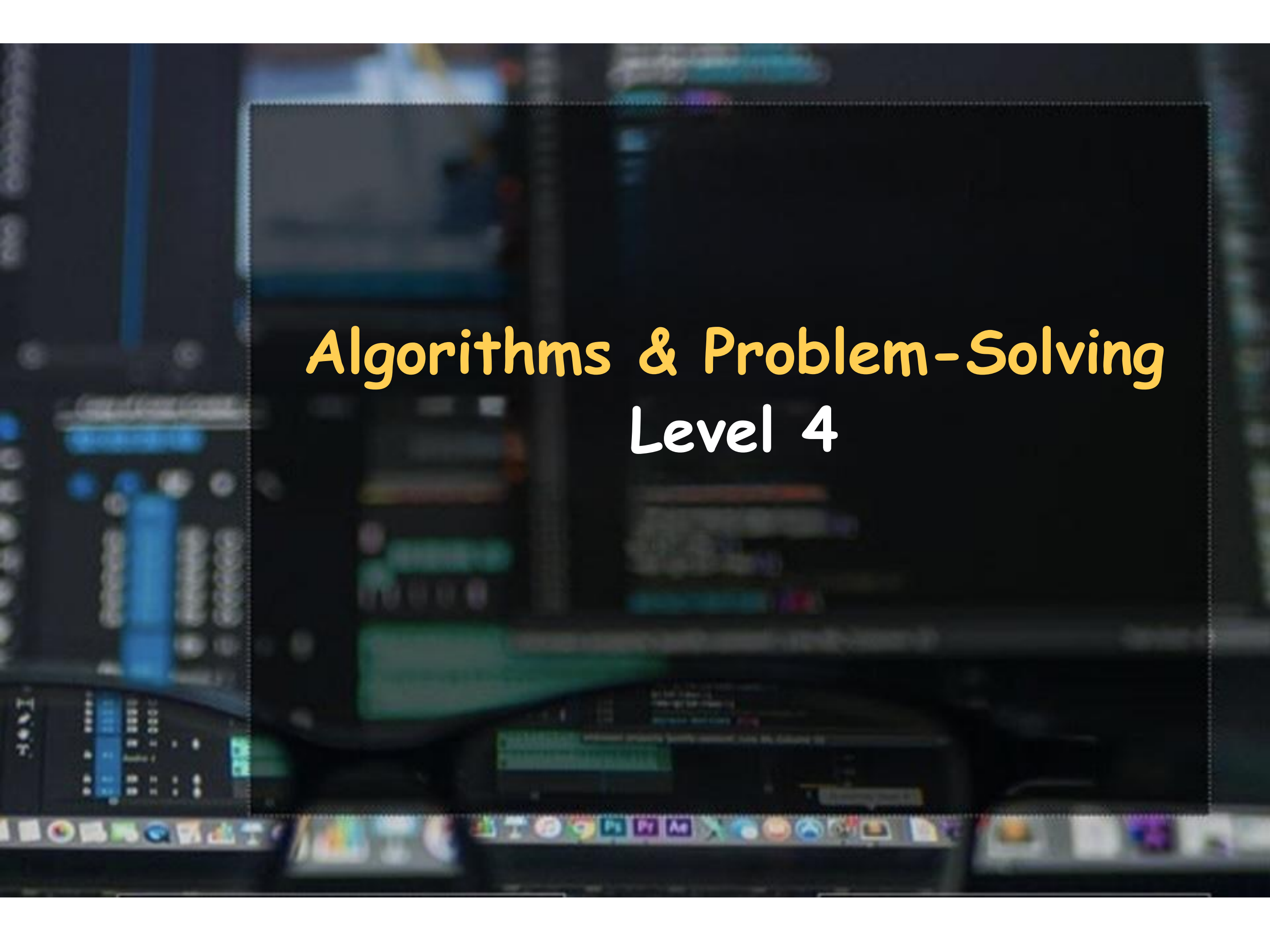
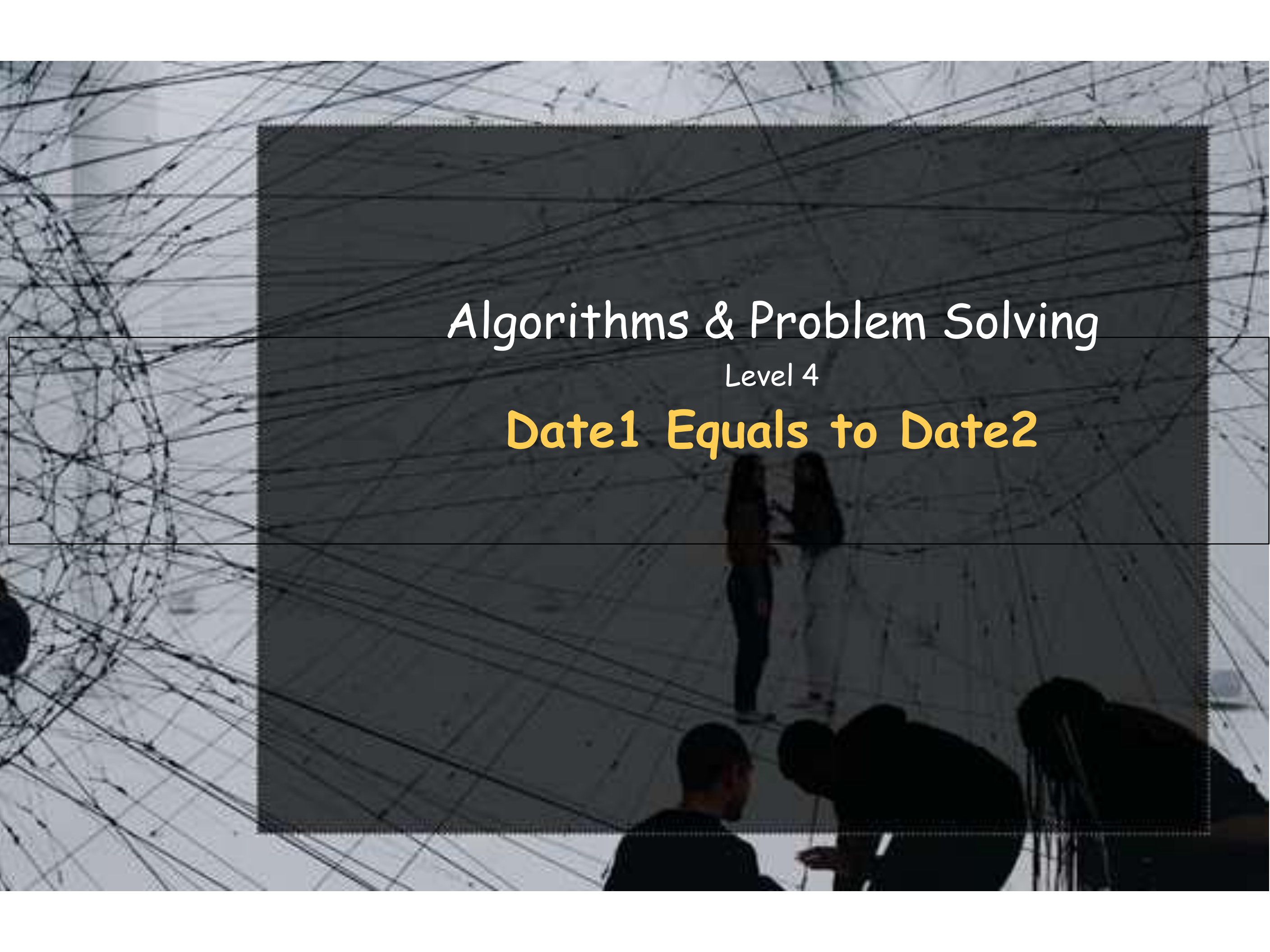




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Date1 Equals to Date2

Definition

- The goal is a program that reads two dates from the user (day, month, year) and then checks if the first date is exactly the same as the second date.



Explanation

- The program uses a struct to group the date components (day, month, year) into a single structure.
- The day, month, and year are each entered via separate functions.
- After the two dates are entered, the program checks if the three values for each are equal:
 - Year
 - Month
 - Day
- If the three values are the same, the two dates are considered equal.

Procedures

- 1 Define the sDate structure containing Year, Month, and Day.
- 2 Create functions to read each of the Day, Month, and Year.
- 3 Create a ReadFullDate() function to read the full date.
- 4 Create an IsDate1EqualDate2() function to compare the two dates.
- 5 In the main() function:
 -  Read date1 and date2
 -  Call the comparison function
 -  Print the appropriate result.



Example:

Please enter a Day? 1

Please enter a Month? 1

Please enter a Year? 2000

Please enter a Day? 1

Please enter a Month? 1

Please enter a Year? 2000

Yes, Date1 is Equal to Date2.

Explanation:

- The comparison is done sequentially:
 - If the year is the same, we move on to comparing the month.
 - If the month is also the same, we check the day.
 - If all are the same → the two dates are the same.



Note:

- The comparison function can be simplified using the following logical expression:
return(date1.Year == date2.Year &&
date1.Month == date2.Month &&
date1.Day == date2.Day);

Summary

- ✓ The program checks if two dates are equal in terms of day, month, and year.
- ✓ It uses structs and structured functions to increase code clarity.

Thank you 😊